

Scientific Writing: Swiss TPH

Lesson 5: Writing the title and acknowledgements

Writing the Methods (and Materials) section

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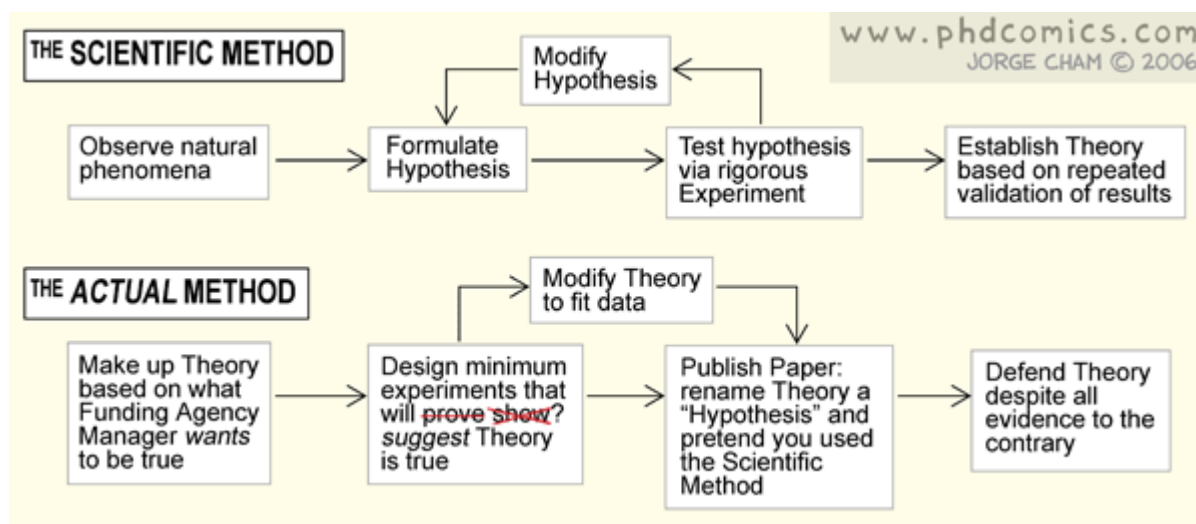


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1. Objectives lesson 5

In this lesson you are going to:

1. Focus on and formulate titles for thesis/article
2. Examine some acknowledgements in articles and theses
3. Examine some Methods sections
4. Explore the contents and structure of the sections
5. Focus on language use in the Methods section, especially the use of tenses and the passive voice
6. Practise some language structures found in this section
7. Learn how to write cohesively and coherently.

2. Writing the title

The title and abstract of a thesis or article give readers a quick overview of your study and help them decide if they want to read the rest of the text. In addition, they form part of the indexing and compilation of reference works for printed and computerized databases. These are some of the reasons why they should accurately reflect the contents of your study and include the necessary key words that would make them easy to find in a database.

Although the title comes first in a thesis and research paper, it is often written last, when one has a better view of the content and structure of the study. Many writers use a working title that is reviewed frequently during the thesis writing process. The following requirements of a good title should be kept in mind (cf. Swales and Feak, 2012: 379):

1. The title should clearly indicate the topic of the study.
2. It should indicate the scope of the study, neither overstating nor understating its significance.
3. It should be self-explanatory to readers in that field of study.

In some cases, writers also include the **nature of the study** in the title, e.g. 'case study', 'report', 'survey', etc.

Some people aim to write striking titles that could also be quite provocative and sometimes contain informal language, but only venture there if you are famous and very respected in the field of study. Until then, rather remain safe!

Titles vary among academic disciplines. Let us investigate some current research titles in your and other research fields.

- What are the requirements for titles at the Swiss TPH, in your department or field of study?
- Perform a search and copy 5 article/thesis titles in your field of study.
- How long are they?
- Do they contain verbs? If so, which?
- How are nouns used?
- Do the titles contain colons? If so, is the section before or after the colon longer? What do the two sections contain?
- Do some titles contain finite verbs?
- Are there any titles that contain direct questions?

- Compare the titles that you found with those of your colleagues. How are they similar? How are they different?
- Can you see any trends in the structure of the titles?

Consider the following **list of thesis titles** from various fields of study below.



Can you detect similarities in titles from the same field of study?

Do you think there are trends in the construction of titles from related fields of study?

- a. Air pollution-related illness: effects of particles (Epidemiology and public health)
- b. Stunting and severe stunting among children under 5 years in Nigeria: a multilevel analysis (paediatrics, public health)
- c. Genetic diversity and malaria vaccine design, testing and efficacy: preventing and overcoming 'vaccine resistant malaria' (Epidemiology, parasite immunology)
- d. Social norms and vaccine uptake: College students' COVID vaccination intentions, attitudes and estimated peer norms and comparisons with influenza vaccine (epidemiology, preventive health, public health)
- e. Emotional Reactivity and Children's Representational Models of Attachment Relationships (Psychology)
- f. Rethinking Stress: The Role of Mindsets in Determining the Stress Response (Psychology)
- g. Origins of phrase structure (Linguistics)
- h. Writing the winning thesis or dissertation: A step-by-step guide (Education)
- i. Neural networks in bioprocessing and chemical engineering
- j. Estimating potential evapotranspiration (Civil engineering)
- k. The determinants of corporate borrowing (Economics)
- l. Effects of International Alliances on Air Traffic: A Case Study of US and International Carriers (Economics)
- m. Massaging the figures? Youth unemployment, school leaving ages and state pensions (Economics)
- n. African women: their struggle for economic independence (Anthropology and Economics)
- o. Explaining the Racial Disparity in Sexually Transmitted Infection: Individual-level behaviour, meso-level structure and macro-level outcomes (Sociology)
- p. Bio-Social Determinants of Child and Adult Mortality in South Africa (Sociology)
- q. The Role of State and Markets in the Integration of Complementary and Alternative Medicine into American Health Care: An Institutional-Level Perspective (Sociology)

In **engineering** and related fields, many titles consist mainly of noun phrases and prepositions. It seems the same is true for epidemiology!

In **medical** fields, the titles tend to be similar to engineering, but some contain finite verbs too, such as 'long-term survival and quality of life *justify* cardiac surgery in the very elderly patient' (Swales and Feak, 2012:380).

Titles in several of the other fields seem to follow the **basic title format** that includes a **shorter main title** that generally describes the work and a **longer subtitle** that explains the finer details of the

research; often separated by a **colon**. Many such examples can be found in the list above. Is this title structure common in your field of study? When using this format, the relationship between the information included before and after the colon could be as follows (ibid:383):

Before the colon:	After the colon
Problem:	Solution
General:	Specific
Topic:	Method
Major:	Minor

Another structure that is frequently used in titles is *verb +ing* (Explaining relationships between...)

Some writers recommend that a title should be 10-12 words long, but it is best to check what seems to be common in your field of study. Whatever the length, the title should be understandable on its own, even without the rest of the text.

If your department has specific requirements for the title format, make sure that you adhere to them. When you write an article for publication in a scientific journal, you must adhere to its requirements when deciding on a title.

3. Writing acknowledgements

Most writers include acknowledgements of some sort in their thesis or research papers.

Examine some theses and papers from your field.

- Copy three acknowledgments included by the writers.
- What does the writer do in the acknowledgements? That is, why does he/she acknowledge these people / companies?
- Does the writer use the first person?
- Where are the acknowledgements printed?

The acknowledgements are mostly given just after the title page of a thesis, or else after the Discussion/ Conclusion or at the end of the thesis or paper.

Swales and Feak (2012:388) list some common elements and examples of acknowledgements:

1. Financial support

Support for this work was provided by (sponsor).

This research was partially supported by a grant from (sponsor).

This research was funded by Contract (number) from (sponsor).

2. Thanks

We would like to thank A, B and C for their help...

I wish to thank A for his encouragement and guidance throughout this project.

We are indebted to B for ... (Note pronunciation of 'indebted'!)

We are also grateful to D for...

3. Disclaimers

However, the opinions expressed here do not necessarily reflect the policy of (sponsor).

The interpretations in this thesis/paper remain my own.

None, however, is responsible for any remaining errors.

However, any mistakes that remain are my own.

The acknowledgements are mostly written in the first person (I/we).

See the next page for 'Writing the Methods Section'.

4. Writing the Methods section

In the Natural Sciences, the methods and materials section of a study is often written first, as it can be based on the research protocols of the study that are followed in the laboratories. In this section it is very important to be clear and concise and to include all the necessary information. As some of you in the class are not doing experiments as part of your research, but all have a way of collecting data, consider what is usually included to describe the way the study was conducted in a thesis or an article such as yours. The Methods section tells the reader how the study was conducted and lead to the results you are publishing.

Many scientific writing books (e.g., Katz 2006; Day and Gastel 2006) recommend giving readers enough detail that they could repeat your work and verify the results themselves. However, studies few published papers come close to this level of detail. Rather than viewing the Methods as the section that allows others to replicate your research, the Methods section is best seen as establishing the credibility of your approach, and thus giving readers a reason to believe your findings. In other words, your article will be seen as authentic and credible science is the Methods section is written well. If the Methods section is not well written and critically read, it can give rise to scientific fraud that is sometimes only discovered long after the initial study had been published.

Consider the article or thesis that you brought with you.

- Is there a methods (and materials) section? If not, is there another section that describes how the study was conducted? What is it called?
- How long is it and what does it contain?
- Are there subheadings in the section? What are they?

The methods (and materials) section describes precisely and in detail everything that was done and used (materials, equipment) to conduct the experiment in order to test the hypothesis or answer the research question. In addition, the reason why a specific method was used is usually given: that is the **justification** for the use of that specific method. In qualitative research, the section might look a bit different, as this type of research is characterised by its aims, which relate to understanding some aspect of social life, and its methods which often generate words, rather than numbers, as data for analysis. Subheadings work well to orientate the reader. A three-part presentation often works, including the background/justification of the methods used, experiments or observations, and analysis of data. If this section has subheadings (which is recommended), they might include Participants, Inclusion/Exclusion Criteria, Sampling, Organisms studied, Materials, Design, and Procedure, etc.

4.1 Outline and content of the Methods section

It would help you to write an **outline** of the Methods section and then fill it in. The outline could look something like this (Adapted from Murray, 2002:158):

1. What did you plan to/ set out to do?
2. How did you plan to do that?
3. Why did you choose that specific approach/ method?
4. What was your hypothesis? / What were your research questions?
5. How did your method fit the questions?

6. Which topics do you need to cover to explain your methods (eg. subjects, treatment groups, etc)?

Annesley (2010:898) suggests answering all the questions in this table that apply to your study:

Who, what, when, where, how, and why questions to consider when writing the Methods section.	
Who	Who maintained the records? Who reviewed the data? Who collected the specimens? Who enrolled the study participants? Who supplied the reagents? Who made the primary diagnosis? Who did the statistical analyses? Who reviewed the protocol for ethics approval? Who provided the funding?
What	What reagents, methods, and instruments were used? What type of study was it? What were the inclusion and exclusion criteria for enrolling study participants? What protocol was followed? What treatments were given? What endpoints were measured? What data transformation was performed? What statistical software package was used? What was the cut-off for statistical significance? What control studies were performed? What validation experiments were performed?
When	When were specimens collected? When were the analyses performed? When was the study initiated? When was the study terminated? When were the diagnoses made?
Where	Where were the records kept? Where were the specimens analyzed? Where were the study participants enrolled? Where was the study performed?
How	How were samples collected, processed, and stored? How many replicates were performed? How was the data reported? How were the study participants selected? How were patients recruited? How was the sample size determined? How were study participants assigned to groups? How was response measured? How were endpoints measured? How were control and disease groups defined?
Why	Why was a species chosen (mice vs rats)? Why was a selected analytical method chosen? Why was a selected experiment performed? Why were experiments done in a certain order?

Peacock (2011, in Swales and Feak, 2012:291) analysed 36 papers from each of the fields of Biology, Chemistry, Physics, Environmental Science, Business, Language and Linguistics, Law, and Public and Social Administration, and proposed the existence of **seven 'Moves'** (a stretch of communication with a specific communicative function) in the methods section:

1. **Overview:** a short summary of the research method at or near the beginning of the section
2. **Research aims, questions or hypotheses:** a description of the aims, research questions or hypotheses
3. **Subjects and/or materials:** in Business, Language and Linguistics, Law, and Public and Social Administration, Sociology, Psychology, Anthropology, amongst others, a description of the study population and sometimes of the corpus; in the sciences, a description of the materials, equipment, ethical issues, and so on.
4. **Location:** a description of where the research took place, and why.

5. **Procedure:** a discussion of the process used to obtain the data that was collected.
6. **Limitations:** a focus on the shortcoming(s) of the method, possibly accompanied by an explanation
7. **Data analysis:** a description of how the data was analysed.

Examine the methods section of the research paper/thesis that you brought with you.

- Can you **identify some of the 'Moves'** listed above? Remember that they do not necessarily appear in the same order as they are given above. Compare your findings with that of your classmates. How do their methods sections differ from yours in terms of the 'Moves' included?
- Do you think the level of detail included is sufficient? Is there enough information for the study to be **replicated** by others?
- How is the information organized? Would it help the reader to include any subheadings?
- What verb tense dominates? Why did the author(s) choose this tense? Could another tense have been used?
- Does active or passive voice dominate? Why?
- Is justification given for the methods that were used? Why and when does one have to justify the methods?
- Does the study include references to the limitations of the methods used? How are the limitations expressed?

In **Qualitative Research**, for example in the fields of Psychology, Sociology, Ethnology, etc. the description of the subjects and location might be quite elaborate. In addition, there might be more explicit information given about the researcher, his/her past experience with the topic, his/her possible role as participant observer, the setting and the subjects. Information might be included about how access to the subjects was gained, about their anonymity, and about other sensitive ethical issues in the study. One also has to indicate the various types of data collection used (interviews, observations, focus groups, document analysis, etc.) and the process of 'reduction' or interpretation that will be used to organize the data. There could also be a discussion of the study's internal validity and its limitations to generalizability.

Consider the following shortened version of a student's methods section in a qualitative study in the field of Sociology:

The aim of this section is to give an overview of the methodology I used in my thesis. In the previous chapter I have already stated that the primary focus of my study is to illustrate the peace interpretations as well as methods of conflict transformation of the X tribe of Native American Indians who live in Y.

In the study I enquire after the meaning of their peace concept and analyse how they deal with conflicts as well as which methods of conflict transformation they usually use. Therefore I turn from the definition of 'what is peace' to 'how is peace interpreted' in a cultural context.

Thus, the principles of qualitative methods of empirical research are in agreement with my research questions, since qualitative methods investigate the 'why' and 'how' and not only the 'what' in the process of data collection and analysis.

Moreover, the qualitative approach propagates a holistic, open view on case studies, allowing and asking explicitly for diversity.

In my field study in Y the spectrum of data collection covers participatory observation, group

discussions and ethnographic field interviews. In combination with participatory observation, group discussions can be used to investigate the pattern of culture as well as modes of interaction. Therefore, I use group discussions for both parts of my research question, as I defined culture as a group pattern, a belief or thought that persists over time and that is shared and not just exercised by one individual.

The field interviews are conducted in such a way that they resemble spontaneous and open conversations rather than formal interviews and for that reason they are closer to the reality of life in the X tribe.

The collected data will be analysed and interpreted using the Documentary Method, which targets the analysis of cultural and collective identity. Hence, the Documentary Method is especially suitable for the interpretation of group discussions, since the construction of knowledge is understood as a collective process in the tradition of Karl Mannheim (Loos/Schäffer 2001).

Since I share a linguistic and socialized culture that is different from that of the research subjects, I will discuss the issues surrounding 'cultural foreignness' (Cappai 2008) and its effects on data analysis in detail in the next section.

Compare the above student's methods section to the methods section in the thesis or paper from your field of study that you brought to the class.

- What similarities are there?
- What differences are there? Highlight them. Consider and describe a wide array of possible differences, such as in language use, type of methods, etc.

In the Social Sciences, this section might take the form of a **narrative** describing the development of a process or a view over time. The use of the first person (I/we) in such a narrative is more common in these disciplines than in the natural sciences.

In quantitative as well as qualitative studies, the methods section could include information about **ethical issues**; two key issues that should be addressed in studies with a human population are **consent** and **confidentiality**.

5. Language use in the Methods section

Consider the following extract from the methods section of a Botany paper (adapted from Naczi, Reznick and Ford, 1998:435, in Swales and Feak, 2012:296), this time focusing on the **language used**:

To detect groups among the specimens and extract the variables that best diagnose the groups, principal components analysis (PCA) was used. Before conducting the analysis, all measurements were standardized so that each variable would have a mean of 0 and a standard deviation of 1. For the PCA, only continuous characters were included. In order to avoid weighting characters, we excluded characters that are probably genetically redundant, as revealed by high values for the Pearson correlation coefficient between all possible pairs of characters.

1. How does the first sentence begin? Is that a pattern that is followed? Highlight patterns.
2. Is there any time indication? Highlight it.
3. Do the authors refer to themselves? How?
4. What tense is mainly used?
5. Is the passive voice used?

Focus on the added **bold**, *italics* and underlining to focus on the language features:

To **detect groups among the specimens and extract the variables that best diagnose the groups**, principal components analysis (PCA) *was used*. Before conducting the analysis, all measurements *were standardized* so that each variable would have a mean of 0 and a standard deviation of 1. For the PCA, only continuous characters *were included*. **In order to avoid weighting characters**, we *excluded* characters that are probably genetically redundant, as revealed by high values for the Pearson correlation coefficient between all possible pairs of characters.

Note that the *we* form as well as the passive voice was used. There is a time indication Before conducting the analysis and a clarifying phrase For the PCA at the opening of the 3rd sentence. Very important is the use of the purpose statements **To detect groups ...** and **In order to avoid...** that explain and justify the chosen procedures. As PCA and the Pearson correlation coefficient are well-known methods, no explanation was given, but if the methods are less well known, one has to justify their use.

5.1 Tense use

The Methods section should be written in the **past tense** because you are describing experiments and protocols that you did in the past. Examples:

- *The experiment was performed at room temperature.*
- *We quantified the drug by immunoassay.*
- *Assay correlation was determined by Spearman rank correlation.*
- *We performed a 2-way ANOVA.*
- *Study participants were recruited from the blood donor service.*
- *Human embryonic kidney cells were cultured in Dulbecco's modified Eagle's medium.*
- *Tissue release of C-reactive protein was monitored after blood flow restriction.*

The 2 exceptions to the use of the past tense are the presentation of summarized data and the introduction of figures and tables within the Methods section:

- *Our protocol is summarized in Figure 1.*
- *Figure 1 illustrates the steps in the procedure.*
- *The data are summarized as median and interquartile range.*
- *The results from these analyses are presented as relative risk ratios with 95% CIs.*

Consult the Academic Phrasebank for examples of typical language use and phrases that one can use when writing the Methods section of a paper.

5.2 Now consider these **linking phrases** that can be used in the Methods section

1. **Purpose statements** such as the ones in the paragraph we analysed above are very common.
 To isolate the leukocytes, ... was used...
 In an effort to evaluate...
 In order to establish...
 To further test this hypothesis...
 To determine the cost...
 In the interest of obtaining useful data...
2. **Time/Temporal phrases** can be used as linking phrases.
 After incubating for 24 hours, the cells were harvested and coloured with...

During the data collection...
 Prior to collecting this information...
 On arrival on campus, the students...
 In the follow-up phase of the study, we...
 After the interview, subjects were...

3. Causal/connective phrases to justify the methodology

Based on previous reports of HR mutations in APL, we performed direct DNA sequencing analysis.

Based on feedback from the pilot study...

On the basis of the literature review...

Because of privacy issues...

In spite of these issues...

In the light of these unexpected findings...

Such linking or transition phrases help to avoid monotony when using predominantly the passive or active voice, for example:

- *After mixing for 1 minute, we added 7 mL methylene chloride . . .*
- *To evaluate ion suppression, extracts were prepared . . .*
- *Because the cells did not adhere to polypropylene, fibroblasts were grown . . .*
- *Because of its signal-enhancing properties, cinnamic acid was added . . .*

In addition to avoiding monotony and improving the flow of the text, transition phrases can be used to start a new paragraph, introduce a new experiment, or describe for the reader why the experiment was performed.

- Can you find any of these forms in the thesis/paper extracts that you brought?
- Which ones are most frequent?
- Add 3 examples from your texts to the list of phrases above.

5.3 Explanation of grammar points relating to the Methods: past and present tenses

We will now focus on **grammar points** relevant to this section, especially the past tense and passive voice.

Explore the use of the simple past and present perfect tenses

To refer to experiences you had or studies that were done at a definite time in the past, use the **simple past tense**. There is normally a time indication in the past: you can usually answer the question 'when did this happen?'.

- I *participated* in a clinical trial in 2010.
- The results *were published* four months ago.
- At the beginning of the previous century, life expectancy *was* 20 years less than it is now.

The present perfect tense is the tense that connects the past and the present.

Use the **present perfect tense** to

- describe trends over a period of time in the (recent) past
- describe events that began in the past and are still continuing

- describe events that have recently happened and are relevant to the present
- refer to an indefinite time in the past.

Some words that are commonly used with the present perfect tense are: *just, already, so far, up to now, since* and *for, ever* and *never*.

Consider these examples of the use of the present perfect tense and match them with the uses described above:

1. She *has never worked* in a laboratory.
2. He *has just published* his first article.
3. She *has worked/has been working* tirelessly on her thesis since Christmas.
4. Over the past century, life expectancy *has increased* steadily.

In academic writing, the present perfect tense is useful for referring to trends, earlier studies or to generally accepted theories, for example:

- Several studies *have demonstrated* that...
- Immunologists *have proven* that...

You could also use the present perfect to refer to previous parts of your own study, for example:

- The first chapter of this thesis *has outlined* the theoretical background to management theory. This chapter will focus on its application in the field of human resources.

Test yourself on the use of the present perfect and past simple tenses:

Complete each sentence with the present perfect or past simple form of the verb in brackets (adapted from Paterson, 2013:9).

1. New research indicates that scientists _____ (discover) a potential cure for some forms of dementia. The key finding _____ (make) two months ago by a team at the University of Basel.
2. The Language Centre _____ (introduce) several new courses in academic writing.
3. Thousands of Irish _____ (emigrate) to the USA between 1845 and 1869 as a result of the great potato famine. One of them _____ an account of his experiences that _____ (just film).
4. Currently editor of The Weekly Business Digest, Wendy Crozier _____ (have) an interesting career. It _____ (start) in 1976 at the BBC.
5. Research _____ (demonstrate) that people can delay the onset of Type 2 diabetes by losing weight.
6. A million Chinese people _____ (move) to Africa in the past 20 years to take advantage of new business opportunities. More than 250 000 Chinese immigrants _____ (arrive) in South Africa alone since 1994.

5.4 Explanation of grammar points relating to the Methods: Active and Passive voice

Resources on writing differ in their preference for either the passive or the active voice for the Methods section. Either style is acceptable if used properly, but a combination of both provides the reader with some variation in presentation of the experiments while also tending to be more concise. Whatever form you choose, be sure to avoid monotony. Here is an example of a description that uses **both forms**:

We recruited study participants from the blood donor clinic. Each participant was asked to fill out a questionnaire, which we then used to classify each individual. Heart rate, blood pressure, and temperature were evaluated, and a 10-mL blood sample obtained for routine chemistry testing. We compiled this information and created a database of patient characteristics vs laboratory results.

Let us focus on the use of the **passive voice** in academic writing in general. While the passive voice is a feature of certain sections of an academic article, making it more impersonal and formal, it should not be overused, as it is not reader-friendly (it is less easily readable) and can make the writing appear wooden and hard to read. If one uses the passive voice too much, it is not reader-friendly, slows the reader down and can be confusing and wordy.

When one wants to write clear, concise and direct sentences, Williams (1989) recommends using the active voice unless one has a strong reason not to. He gives the following examples to illustrate this point.

- | | |
|---------|---|
| Passive | a. It is believed by the candidate that a ceiling must be placed on the budget by Congress. |
| Active | b. The candidate believes that Congress must place a ceiling on the budget. |
| Passive | c. It was earlier demonstrated that heart attacks can be caused by high stress. |
| Active | d. Brown earlier showed that high stress can cause heart attacks. |

However, there are sometimes good reasons to use the passive voice:

- To emphasize the action rather than the actor.
e. After long debate, the proposal *was endorsed* by the long-range planning committee.
This is the reason why the passive voice is frequently used in the Methods section of a paper.
- To keep the subject and focus consistent throughout a passage.
f. The data processing department recently presented what proved to be a controversial **proposal** to expand its staff. After long debate, the **proposal was endorsed** by...
- To be tactful by not naming the actor.
g. The procedures *were* somehow *misinterpreted*.
- To describe a condition in which the actor is unknown or unimportant.
h. Every year, thousands of people *are diagnosed* as having cancer.
- To create an authoritative tone.
i. Visitors *are not allowed* after 9:00 p.m.

Passive voice practice

Change these sentences from the active to the passive voice (Bailey, 2015:139).

1. We collected the data and compared the two groups.
2. I interviewed 120 people in three social classes.
3. They checked the results and found several errors.
4. We will analyse the findings.
5. He asked four doctors to give their opinions.
6. She wrote the report and distributed ten copies.

5.5 Writing cohesively and coherently

See the slides for more information and examples.

6. Assignment 5A

Assignment 5A Methods

Note: you have a choice between doing Assignment 5A (Methods) and Assignment 5B (Results)

See the full instructions in the 'Instructions for assignments' document.

7. Bibliography

Annesley, Thomas M. 2010. Who, What, When, Where, How, and Why: The Ingredients in the Recipe for a Successful Methods Section. *Clinical Chemistry* 56:6, 897–901.

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Key:

Test yourself on the use of the present perfect and past simple tenses

1. have discovered, was made
2. has introduced
3. emigrated, wrote , has just been filmed
4. has had, started
5. has demonstrated
6. have moved, have arrived

Passive voice exercise

1. The data were collected and the two groups compared. OR The data was collected and the two groups were compared
2. 120 people in three social classes were interviewed.
3. The results were checked and several errors found.
4. The findings will be analysed.
5. Four doctors were asked to give their opinions.
6. The report was written and ten copies were distributed.